

Designing a Model

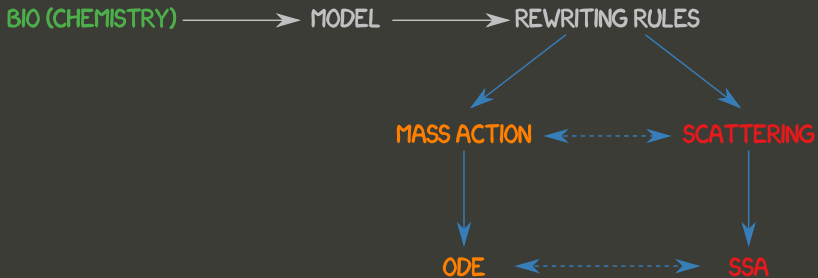
Dario Pescini¹

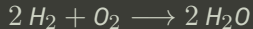
Università degli Studi di Milano-Bicocca

Dipartimento di Statistica e Metodi Quantitativi

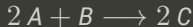
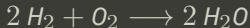
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Design of mechanistic models









Rewriting rules



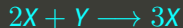
The objects are the **players** and the reactions are their **interactions**.

The elements used to describe the story are:

- the **alphabet** $\{A, B, C\}$
- the set of **rewriting rules** (i.e. reactions) $2 A + B \longrightarrow 2 C$

Rewriting rules

$\{A, B, X, Y, E\}$



- The rewriting rules are a “natural” and well spread representation of biochemical models
- they overlap with the stoichiometric system representation
- they are a **unique and unambiguous representation** for both deterministic and stochastic frameworks

From rewriting rules to matrices

$$A \longrightarrow X$$

$$B + X \longrightarrow Y + D$$

$$2X + Y \longrightarrow 3X$$

$$X \longrightarrow E$$

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$$\alpha_{1,1}x_1 + \alpha_{1,2}x_2 + \cdots + \alpha_{1,N}x_N \xrightarrow{c_1} \beta_{1,1}x_1 + \beta_{1,2}x_2 + \cdots + \beta_{1,N}x_N$$

$$\alpha_{2,1}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{2,N}x_N \xrightarrow{c_2} \beta_{2,1}x_1 + \beta_{2,2}x_2 + \cdots + \beta_{2,N}x_N$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$\alpha_{M,1}x_1 + \alpha_{M,2}x_2 + \cdots + \alpha_{M,N}x_N \xrightarrow{c_M} \beta_{M,1}x_1 + \beta_{M,2}x_2 + \cdots + \beta_{M,N}x_N$$

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⋮

⋮

⋮

$$\alpha_{M,1}x_1 + \alpha_{M,2}x_2 + \cdots + \alpha_{M,N}x_N \xrightarrow{c_M} \beta_{M,1}x_1 + \beta_{M,2}x_2 + \cdots + \beta_{M,N}x_N$$

$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,N} \\ \alpha_{2,1} & \alpha_{2,2} & \cdots & \alpha_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{M,1} & \alpha_{M,2} & \cdots & \alpha_{M,N} \end{pmatrix}$$

$$\begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \cdots & \beta_{1,N} \\ \beta_{2,1} & \beta_{2,2} & \cdots & \beta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M,1} & \beta_{M,2} & \cdots & \beta_{M,N} \end{pmatrix}$$

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{pmatrix}$$

From rewriting rules to matrices

$$L = \begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \dots & \alpha_{1,N} \\ \alpha_{2,1} & \alpha_{2,2} & \dots & \alpha_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{M,1} & \alpha_{M,2} & \dots & \alpha_{M,N} \end{pmatrix}$$

$$R = \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \dots & \beta_{1,N} \\ \beta_{2,1} & \beta_{2,2} & \dots & \beta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M,1} & \beta_{M,2} & \dots & \beta_{M,N} \end{pmatrix}$$

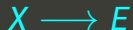
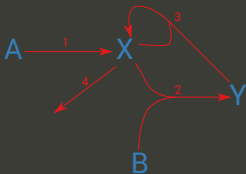
$$\vec{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_M \end{pmatrix}$$

$$D = R - L = \begin{pmatrix} \delta_{1,1} & \delta_{1,2} & \dots & \delta_{1,N} \\ \delta_{2,1} & \delta_{2,2} & \dots & \delta_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_{M,1} & \delta_{M,2} & \dots & \delta_{M,N} \end{pmatrix}$$

the BZ Reaction and the Brusselator



The **Brusselator**(JCP 46 1695) is a theoretical extremely simplified version (though physically unrealistic) of the **BZ** (Belousov-Zhabotinskii) Reaction now days recognized as being the **prototype for Chemical Oscillators**:



From rewriting rules to matrices

$$A \xrightarrow{c_1} X$$

$$B + X \xrightarrow{c_2} Y + D$$

$$2X + Y \xrightarrow{c_3} 3X$$

$$X \xrightarrow{c_4} E$$

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} \xrightarrow{c} = \begin{pmatrix} 1 \\ 5 \cdot 10^{-3} \\ 2.5 \cdot 10^{-5} \\ 1.5 \end{pmatrix}$$

Not only bio-chemistry!

It is possible to exploit rewriting rules **to describe** any kind of transition.

Lotka-Volterra

The Lotka-Volterra model aims to describe predator prey interactions to simulate the populations dynamics.

In this model there are:

Lotka-Volterra

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In this model there are:

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- predators



Lotka-Volterra

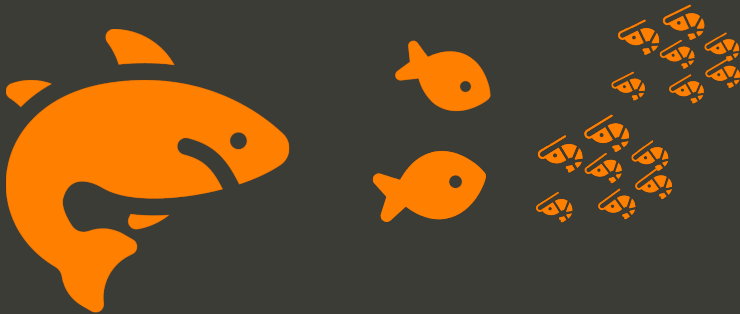
The Lotka-Volterra model aims to describe predator prey interactions to simulate the populations dynamics.

In this model there are:

- preys 
- predators 
- prey's food 

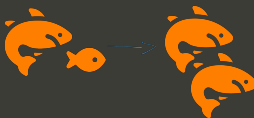
Lotka-Volterra

In a single patch:



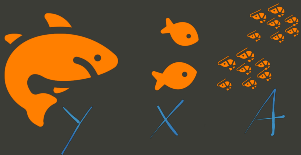
Lotka-Volterra

They interact:



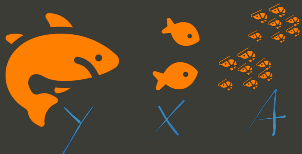
Lotka-Volterra

By associating an alphabet it is possible to design a mathematical model by means of rewriting rules:

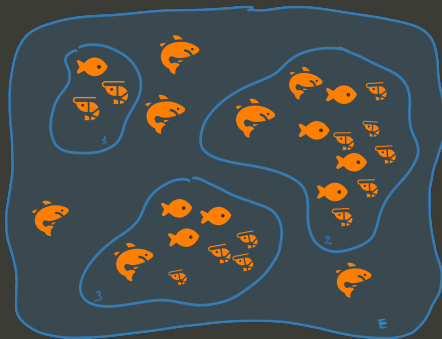


Lotka-Volterra

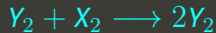
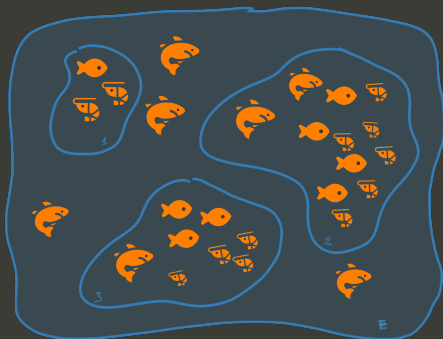
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From Lotka-Volterra to metapopulations



From Lotka-Volterra to metapopulations



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